Reward engineering, simulator fidelity and lap-time benchmarks

RL driving agent in OCD: Barcelona and Shanghai

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Outline

1. Reward engineering: from line tracking to a time-optimal reward
2. Vehicle dynamics: AV-21 $\rightarrow$ `sf70h_tum` $\rightarrow$ `sf70h_f1`, parameter by parameter
3. Verification: Limebeer-style optimal control (OCP) reproduced
4. Results: Barcelona vs. Limebeer, Shanghai vs. real world and QSS
5. Open points and next steps

Reward engineering

Old reward (`ref_tracking`, run `full1`)

Per 50 Hz tick, e = lateral offset to the reference line ξ_{ref}:

$$r = a\,\Delta s \;-\; b\,\Delta t\,\big(1+\underbrace{\dot{u}^{\top} W\,\dot{u}}_{\text{jitter}}\big) \;-\; \underbrace{c\,e^{2}}_{\text{tracking}} \qquad a = b = 1,\; c = 0.1,\; W = \mathrm{diag}(0.01, 0.001, 0.001)$$

Failure (off track, stall, wrong way): $P = 50 + 30\,c\,e_{\text{exit}}^{2}$.

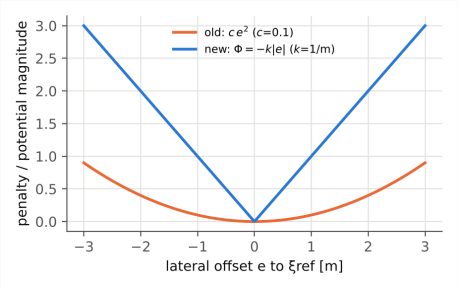
What we saw (Shanghai, AV-21 car, full1 at 2.95M steps):

- Driver glued to ξ_{ref}: mean $|e| = 0.39$ m, max 1.83 m.
- Lap cost split: time 123.3, **tracking 175.9**, smoothness 10.4. Tracking was a larger term than lap time.
- Best clean lap 122.85 s. The reference line is a *safe* min-curvature line, so following it closely means driving slowly.

New reward (`time_pbrs`, R1): what changed

$$r = a\,\Delta s - b\,\Delta t \;+\; \big[\gamma\,\Phi(s') - \Phi(s)\big], \qquad \Phi(s) = -k\,|e(s)|,\; k = 1\,\mathrm{m}^{-1},\; \gamma = 0.992$$

1. **Jitter term removed.** $\dot{u}^\top W \dot{u}$ is no longer rewarded (only logged).
2. **Line penalty linear, not squared:** $k|e|$ instead of $c\,e^2$.
3. **Line term is potential-based shaping** (Ng et al. 1999), so the optimal policy is the same as with no line term at all. ξ_{ref} only guides early learning.
4. Constant failure penalty $P = 100$; a stall truncates the episode instead of ending it.



Why these two changes

Removing the jitter (smoothness) term

- Top racing agents (Linesight/Trackmania, GT Sophy) use no smoothness term in the reward.
- Christmann et al. (IROS 2024): a smoothness penalty can make the agent settle for worse performance, and its weight is hard to tune.
- The time cost already penalises wasted inputs. Smoothness stays a logged diagnostic.

$|e|$ **instead of** e^2

- e^2 is flat near the line and steep far from it. Small deviations cost almost nothing, large ones dominate the return.
- $|e|$ has a constant gradient (Linesight also shapes on a linear distance).
- Inside a PBRS potential the term cannot change the final driver. It only speeds up early learning.

Effect of the new reward

Run	Car / reward	Best clean lap 2	Own QSS (safe line)
full1	AV-21, old reward	122.85 s	133.9 s
r1_warm	AV-21, new reward, warm start	123.04 s	133.9 s
sf70h_mac_r1	sf70h_tum, new reward, from scratch	**103.24 s**	104.78 s
p1_TUM	sf70h_tum + TUM-like setup, new reward	**101.27 s**	103.56 s

- The warm start kept the old line-following habits and gained nothing. From scratch, the free driver **beats the QSS lap on the safe line** by 1.5–2.3 s, because it is no longer tied to that line.
- Cost: lower clean rate. Only 48/194 deterministic evaluations were clean in sf70h_mac_r1.

Vehicle dynamics: what changed in the car

Three cars, built in layers

	`av21` (shipped OCD)	`sf70h_tum` (27 Oct)	`sf70h_f1` (1 Nov)
Idea	Dallara AV-21 (Indy Autonomous Challenge)	AV-21 model with every value TUM's `F1_Shanghai.ini` gives	`sf70h_tum` + F1 values from papers and the FIA rules
Mass, geometry, aero	AV-21	TUM	TUM
Tyre peak μ	AV-21 (1.46 / 1.16)	TUM (1.65–2.15)	TUM
Tyre slip stiffness, shape	AV-21	AV-21	PL14 (Limebeer)
Steering lock	0.6 rad	0.6 rad	0.30 rad
Heave springs	off	off	1.0 / 0.25 MN/m
Brakes	12 kN m, 0.50	12–15 kN m, 0.50	15 kN m, 0.58
Footprint	4.2×2.0 m	4.6×2.0 m	5.006×2.07 m

Each new car is opt-in. `av21` and `sf70h_tum` stay bit-identical (Shanghai `lap.csv` sha256 unchanged, 85 unit tests). There are also two variants used only for diagnosis: `sf70h_tum_debug` (31 Oct) and `sf70h_f1_iz560`.

`sf70h_tum` (1/3): chassis, mass, aero

Parameter	AV-21	`sf70h_tum` (from TUM)
Mass (driver, no fuel)	800 kg	733 kg (`general.m`)
Wheelbase	3.2 m	3.6 m (l_f 1.968 / l_r 1.632)
CoG height	0.30 m	0.335 m
Track F / R	1.6 / 1.5 m	1.6 / 1.6 m
Rolling radius	0.30 m	0.3299 m ($2.073/2\pi$)
Air density	1.225	1.18 kg/m^3
Rolling resistance	0.025	0.03
C_zA F / R, C_xA [m^2]	2.20 / 2.68, 1.56 (already TUM)	
$I_x/I_y/I_z$ [kg m^2]	100 / 500 / 1000	97.6 / 579.8 / 1159.6

Inertias are scaled, not sourced (PLACEHOLDER):

$$I_{y,z} \propto mL^2, \qquad I_x \propto m\,t^2$$

(L wheelbase, t mean track), starting from the AV-21 values.

Moving the CoG back (45.3 % front vs 46.1 %) with the same aero balance, plus 3.7 % less downforce from the lower ρ, makes this step *slightly slower* in the QSS.

sf70h_tum (2/3): tyre peak friction, mapped exactly

TUM uses a tyre that is linear in load. So does MF5.2's peak friction:

$$\mu_{\text{TUM}}(F_z) = \mu_0 + \tfrac{d\mu}{dF_z}(F_z - f_{z0}), \qquad \mu_{\text{MF}}(F_z) = (P_{D1} + P_{D2}\, df_z)\,\lambda_\mu, \quad df_z = \tfrac{F_z - F_{\text{nom}}}{F_{\text{nom}}}$$

Both use $F_{\text{nom}} = f_{z0} = 3000\,\text{N}$, so $P_{D1} = \mu_0/\lambda_\mu$ and $P_{D2} = \tfrac{d\mu}{dF_z} f_{z0}/\lambda_\mu$, with $\lambda_\mu = 0.97$ kept.

	μ_0	P_{D1} new	P_{D1} AV-21	P_{D2}
Front x	1.65	1.701	1.50	−0.155
Front y	1.85	1.907	1.20	−0.155
Rear x	1.95	2.010	1.50	−0.155
Rear y	2.15	2.216	1.20	−0.155

AV-21 P_{D2}: -0.04 (x), -0.09 (y).

- Peak μ_x matches TUM *exactly* at every load. Peak μ_y matches on average ($\pm 2\,\%$ split from the AV-21 zero-slip offset PVY1).
- Grip at 3 kN: $+34\,\%$ longitudinal (rear), $+79\,\%$ lateral (rear) vs the AV-21.
- Only these 4 numbers per wheel changed. Everything else in the tyre stayed AV-21.

sf70h_tum (3/3): powertrain, ERS, DRS, brakes, QSS

ERS (new in `powertrain.py`, from TUM)

- 120 kW after losses, $\eta = 0.9$, ≤ 200 N m at the crank, only above 27.8 m/s
- 4 MJ per lap, "first come, first boost". No recuperation (TUM has it: -1.44 s)

DRS: QSS only, $C_xA \times 0.83$ in TUM's two Shanghai zones while accelerating.

QSS: friction *ellipse* $(a_x/a_{x,\max})^2 + (a_y/a_{y,\max})^2 \leq 1$ (TUM's exponent 2). The AV-21 used a diamond (exponent 1), which is 7.8 s slower on the same g-g-v. The g-g-v is now measured up to 90 m/s.

RL runs: **ERS and DRS off**.

Brakes: torque at which the first axle reaches its peak (bias 0.50, quasi-static)

Speed	AV-21	sf70h_tum
0 m/s	2.9 kN m	3.8 kN m
60 m/s	6.7	8.7
80 m/s	9.5	12.3
90 m/s	11.2	**14.3**

The old 12 kN m could not lock the F1 car above ≈80 m/s, so above that speed the brakes, not the tyres, limited deceleration. Raised to **15 kN m**.

What each layer is worth (QSS, Shanghai, safe line, `Setup()`)

Step	Lap [s]	$v_{\max}$ [km/h]
AV-21 (exp. 1)	133.92	288
AV-21 (exp. 2)	126.43	288
+ TUM tyre μ	106.42	288
+ mass 733 kg	104.48	288
+ geometry	104.78	301
+ ERS	103.25	311
+ DRS	102.82	321

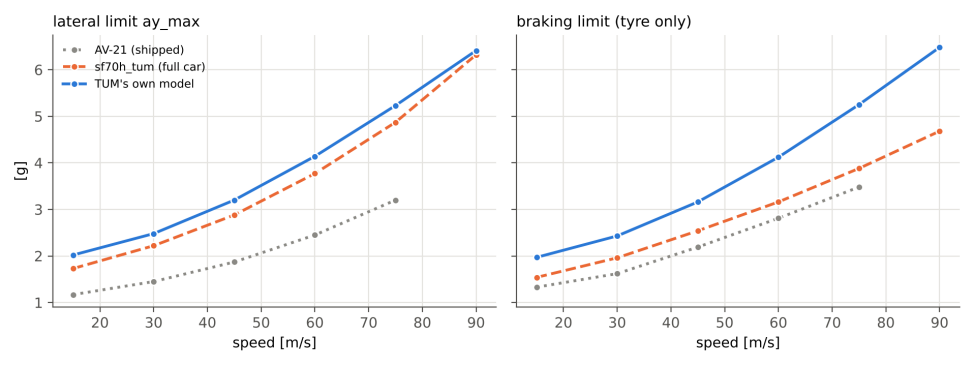
Exponent 2 from "+ TUM tyre μ" on. Peak friction alone is worth ≈20 s.

Against TUM's laptime-sim, **same raceline**, ERS/DRS off: ours 96.86 s, TUM 94.89 s (+2.0 s). That splits into:

- **car limits** +5.0 **s**: our g-g-v is lower (next slide)
- **solver/powertrain** −3.1 **s**: we omit TUM's rotating-mass factor, rolling resistance under acceleration, 0.5 m/s cornering margin and $r(v)$

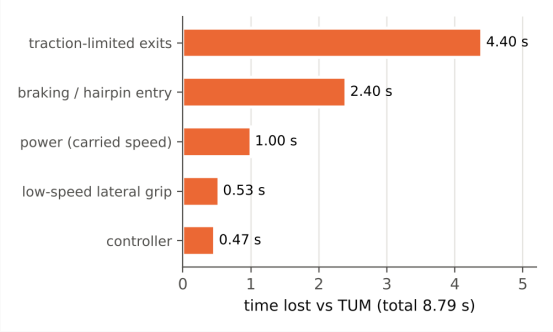
The safe line costs ≈8 s against the raceline (104.78 vs 96.86 s): it is 48 m longer and built with a 4.5 m width allowance.

Measured g-g-v limits vs TUM's own model



Lateral is close to TUM (4.87 vs 5.23 g at 75 m/s). **Braking is the biggest shortfall** (3.88 vs 5.25 g). Causes: the brake ramp is labelled at its start speed (the peak is reached 15–20 m/s lower), bias 0.50 vs TUM's ideal split, pitch/slip transients during the 0.6 s ramp, and a 61 % front roll-stiffness split (AV-21 default) vs TUM's 45 % static split (0.45 gains 0.22 s).

Diagnosis: TUM's 94.9 s lap driven closed-loop in OCD



- TUM-aligned debug car: constant aero, TUM's $r(v)$, CoG-only track check. A 50 Hz controller tracks TUM's line and speed (feed-forward + PI, TC at 8 % slip, ABS).
- Driven **102.99 s** vs QSS 95.73 s vs TUM 94.89 s. The *quasi-static* limits match; the time is lost in *transients*.
- Longitudinal slip stiffness $\times 1.6$: -2.69 **s**, QSS unchanged.
- Artefact: a locked LSD makes the rear wheels fight (3.8 kN Fx difference), but an open LSD is worse ($+2.36$ s).

Why the AV-21 slip stiffness hurts: the MF5.2 algebra

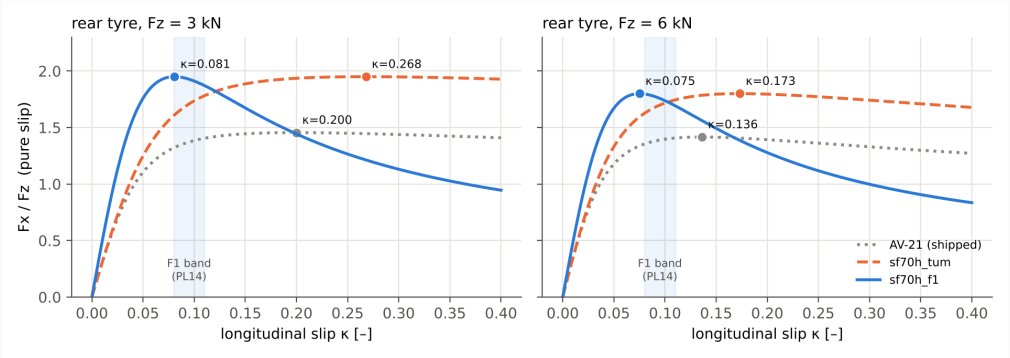
$$F_x = D\sin\big(C\arctan(B\kappa - E(B\kappa - \arctan B\kappa))\big), \qquad D = \mu F_z,$$

$$K_x = F_z(p_{K1} + p_{K2}\,df_z)\,e^{p_{K3}df_z}, \qquad B = \frac{K_x}{CD}$$

- The initial slope K_x is set by PKX only. The peak sits at a fixed $B\kappa$ (for $E = 0$, $B\kappa^* = \tan\frac{\pi}{2C}$), so $\kappa^* \propto \dfrac{\mu}{K_x/F_z}$.
- `sf70h_tum` raised μ by 34–79 % and left K_x alone, so the peak **moved out to larger slip**. On top of that, the AV-21 curvature $E_x = 0.7$ flattens the top of the curve.
- AV-21 K_x/F_z = 29.3 / 30.7 / 35.3 at 2 / 3 / 6 kN. F1 (PL14) = 43.7 / 42.5 / 38.5: the AV-21 tyre is **25–33 % softer** at low load.

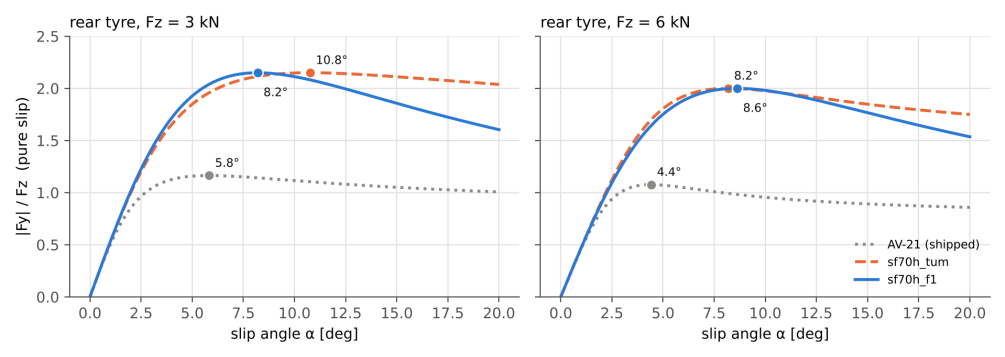
Rear tyre, peak slip	AV-21	`sf70h_tum`	`sf70h_f1`
κ^* at 3 / 6 / 12 kN	0.20 / 0.14 / –	0.27 / 0.17 / 0.10	0.081 / 0.075 / 0.064
α^* at 3 / 6 / 12 kN	5.8 / 4.4 / –°	10.8 / 8.2 / 7.8°	8.2 / 8.6 / 10.8°

Longitudinal force: where the peak is



Same peak μ for `sf70h_tum` and `sf70h_f1`, but the F1 tyre reaches it at $\kappa \approx 0.08$, inside the F1 band (PL14: 0.08–0.11). `sf70h_tum` needs 0.17–0.27, and drops off slowly after the peak. Curves: OCD's MF5.2 equations in Python; they reproduce the committed sweep exactly.

Lateral force: shape changed, stiffness kept



PKY (cornering stiffness) is unchanged: the AV-21 K_y is already within 4 % of PL14 at 2–6 kN. Only the shape changed (C_y 1.5 → 1.9, E_y → 0). So the peak is sharper and sits at 8–9° (it was 10.8° at 3 kN), and the force falls faster after the peak.

How the F1 tyre parameters were derived (option B, approved 1 Nov)

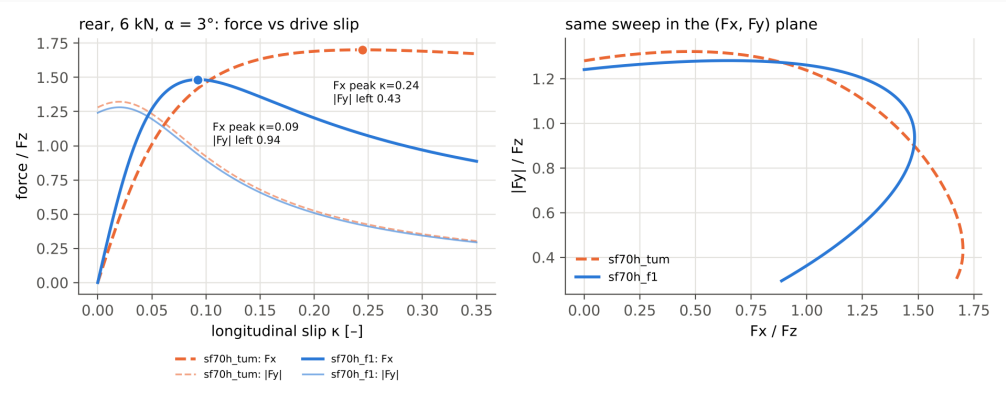
PL14 tyre (Perantoni & Limebeer 2014, Table 3):

$$\mu = \mu_{\max} \sin\big(Q \arctan(S\rho)\big), \quad S = \frac{\pi}{2 \arctan Q} = 1.446, \quad \rho = \sqrt{\kappa_n^2 + \alpha_n^2}, \quad \kappa_n = \kappa/\kappa_{\max}$$

1. This is MF with $C = Q = 1.9$, $E = 0$, $B = S/\kappa_{\max}$, so $\dfrac{K_x}{F_z} = Q\,S\,\dfrac{\mu}{\kappa_{\max}} = 2.747\,\dfrac{\mu_{x,\mathrm{TUM}}(F_z)}{\kappa_{\max}(F_z)}$.
2. $\kappa_{\max}$ is linear in load: 0.11 at 2 kN, 0.10 at 6 kN. μ stays TUM's.
3. Fit $K_x/F_z = p_{K1} + p_{K2}\,df_z$ exactly at 3 and 6 kN, with $p_{K3} = 0$:
   front $p_{K1} = 42.17$, $p_{K2} = -0.96$; rear $p_{K1} = 49.84$, $p_{K2} = -0.38$ (AV-21: 30.7, 0.27, 0.13).
4. PCX1 = PCY1 = 1.9; all PEX_{1-4}, $\mathrm{PEY}_{1-5} = 0$.

Peak at $\rho = \tan(\pi/2Q)/S = 0.751$, i.e. $\kappa^* \approx 0.75\,\kappa_{\max} \approx 0.08$.
Option A (PL14's stiffness directly: $p_{K1} = 42.5$, $p_{K2} = -4.0$) would keep PL14's peak slips only with PL14's lower μ.

Combined slip: what a corner exit looks like



At $\alpha = 3°$, `sf70h_tum` reaches its Fx peak only at $\kappa = 0.24$, with 0.43 of lateral force left. The F1 tyre peaks at $\kappa = 0.09$ with 0.94 left. Caveat: the combined-slip weights (RBY1 = 20.6, ...) are still the AV-21's in both cars. That is why lateral force falls ≈30 % by $\kappa = 0.1$, which is harsher than TUM's friction ellipse.

`sf70h_f1`: the non-tyre changes

- **Steering lock** 0.6 → **0.30 rad** road-wheel angle. Source: "Normally an F1 car has a steering lock of 17°" (Monaco spec 21–22°). The OCP needs at most 0.27 rad (T14), so the lock does not bind. Rate 1.5 rad/s and T_{PT1} 0.06 s kept.
- **Heave springs** (OCD patch 0001, previously off). Per-wheel rate $k = m_a(2\pi f)^2$, with m_a the corner sprung mass:
  front **1.0 MN/m** = 12.95 Hz (corner spring 107 kN/m, so heave is 9× stiffer); rear **0.25 MN/m** = 5.85 Hz (corner 83 kN/m). The front knob range was widened from 0–8 to 0–16 Hz. Values derived from Masouleh & Limebeer 2016.
- **Brake bias** default 0.50 → **0.58** (ideal 0.58–0.60 for this car). **Torque** 15 kN m car default (14.3 needed at 90 m/s).
- **Footprint** for the track check: 4.6 × 2.0 → **5.006 × 2.07 m**, $x \in [-2.342, 2.664]$ m, $y = \pm 1.035$ m.
  Built from: FIA 2017 rear overhang ≤ 0.71 m, wheelbase 3.594 m, length 5.0 m, track/2 + half the 0.47 m rear wheel.
- I_z: kept at 1159.6. Variant `iz560` uses PL14's 450 kg m^2 scaled by mL^2, which gives 560.3.

New finding: the F1 car porpoises

Mechanism

- Stiff heave springs with `heave_damping_ratio` $= 0$ (the default) leave a lightly damped **4.5 Hz heave mode**.
- The PLACEHOLDER aero map raises C_zA by **2 % per cm** below the reference ride height. As the car drops, downforce grows and pushes it down further.
- The effect scales with v^2, so the oscillation grows with speed. OCD has no floor contact to stop it.

Evidence

- Front ride-height std at 90 m/s: **9.8 mm**, vs 3.0 mm for `sf70h_tum`.
- Not a numerical artefact: 11.25 mm at a 2 ms step vs 10.80 mm at 0.5 ms.
- Heave damping 0.3 (diagnostic) cuts it $\approx 10\times$.
- Barcelona: the 90 m/s settle collapses (wheel lift, wheelspin), so the g-g-v is only valid up to 75 m/s.

Decision needed: add a heave-damping default for this car (no sourced value exists), or soften the heave springs.

Effect of sf70h_f1: handling and lap time

Ramp steer (p1_TUM knobs)

v [m/s]	$a_{y,\max}$ [g] tum / f1	US grad. [deg/g]
30	2.29 / 2.29	0.28 / 0.29
45	3.04 / 3.05	0.22 / 0.25
60	4.03 / 3.99	0.17 / 0.18
75	5.15 / 5.10	0.12 / 0.07

Same peak grip and understeer gradient. The F1 tyre is sharper past the peak.

Lap time (15 kN m, ERS off)

	sf70h_tum	sf70h_f1
QSS Shanghai safe line	103.56	102.40
QSS Shanghai raceline	95.65	94.67
QSS Barcelona safe line	82.89	82.44
OCP Shanghai	94.25	94.62
OCP Barcelona	78.76	79.38
Closed-loop Shanghai	102.87	101.84

QSS gains about 1 s, mostly from the 0.58 bias. The OCP is 0.4–0.6 s slower because of the wider car (narrower corridor) and the wider lateral peak.

The controller has to match the tyre

Closed-loop TUM lap in OCD, `sf70h_f1`, lap 2 at $f = 1.00$:

Controller / car	Lap 2 [s]
Old controller (TC target 0.08 = new Fx peak)	spin
TC target 0.04 / 0.05 / **0.06** / 0.07	103.86 / 102.65 / **101.84** / spin
+ heave damping 0.3	101.29
`iz560` + TC 0.06 + heave damping 0.3	100.72
`sf70h_tum`, original controller	102.87

- The TC target alone moves the lap by ≈2 s, as much as the car change itself.
- **For RL on** `sf70h_f1`**:** the env scales steering to 0.6 rad and must be changed to 0.30. The demo driver's TC targets need re-tuning for the peakier tyre.
- I_z effect is small and depends on the method: QSS +0.10/+0.23 s, closed loop −0.4 to −0.6 s, OCP +0.05 to +0.08 s.

Still AV-21 / PLACEHOLDER after `sf70h_f1`

No F1 source found

- I_x, I_y (scaled); wheel masses 15 kg and spin inertias 1.15 / 1.35 kg m^2
- Combined-slip weights R*; offsets PVY/PHY/RVY ($\pm 2\,\%$ μ_y split)
- Relaxation lengths 0.3 / 0.2 m (hard-coded in OCD C++)
- Roll / pitch centres 0.2 m; dampers, heave damping, packers
- Steering rate 1.5 rad/s, T_{PT1} 0.06 s

Known modelling gaps

- Aero map: C_zA $\times$0.6–1.4 with aero_level, 2 %/cm ride height (PLACEHOLDER until measured in Assetto Corsa)
- Tyre vertical stiffness 250 kN/m (F1 estimate 266–285)
- No speed growth of the tyre radius (TUM: +4.5 % per 200 km/h)
- No ERS recuperation; DRS only in the QSS
- OCD's LSD cannot represent TUM's ideal LSD
- Static front ride height at $v_{\max}$ is already negative (no floor contact)

Verification: Limebeer-style OCP

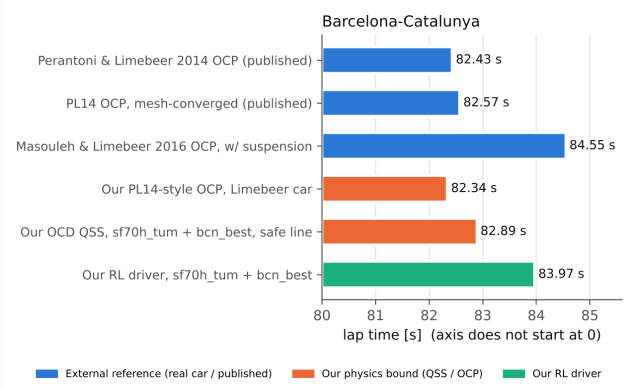
Minimum-lap-time OCP, as in Perantoni & Limebeer 2014

- Planar body, quasi-static load transfer, 4 combined-slip tyres, LSD, aero, power limit. Distance-domain direct collocation (CasADi + IPOPT), mesh down to 0.5 m.
- **D-check, Barcelona with Limebeer's car: 82.34 s** vs. PL14 mesh-converged 82.57 s, i.e. $-0.28\,\%$ (pass band $\pm 1.5\,\%$).
- Mesh-converged within 0.005 s. KKT residuals $< 5 \times 10^{-7}$. IDAS re-integration agrees to < 1 ms in lap time.
- Same model with the SF70H-TUM values: **Shanghai 94.25 s** (TUM QSS 94.89 s), **Barcelona 78.76 s**. With the `sf70h_f1` tyre: 94.62 / 79.38 s.

Limits: local optima only (a different initial guess gave $+0.55$ s). No suspension and no wheel-spin dynamics. This is an external reference, not an OCD lap.

Results

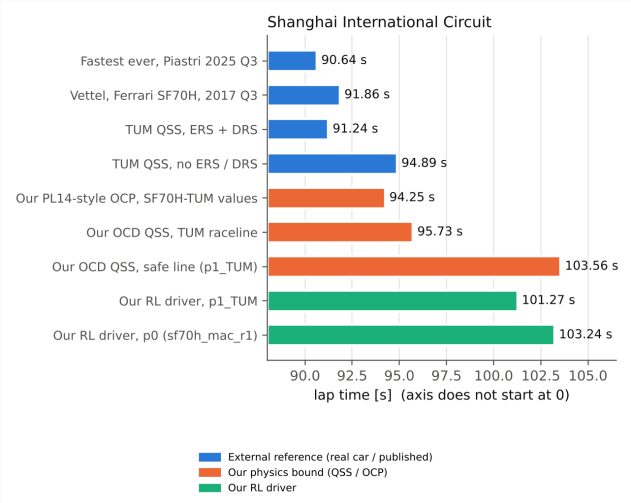
Barcelona: Limebeer's OCP vs. our RL driver



- RL: **83.97 s** (step 5.05M of 7.05M).
- +1.40 s (+1.7 %) to PL14's 82.57 s.
- +1.08 s (+1.3 %) to our own QSS (82.89 s).
- Faster than ML16's suspension model (84.55 s).

Not the same car: PL14 uses a 2013-era F1 car (660 kg, C_L 3.0, 560 kW). Ours is `sf70h_tum` (733 kg, 575 kW) with `bcn_best` at aero_level 1.0 on a PLACEHOLDER aero map. Our OCP with SF70H-TUM values gives 78.76 s.

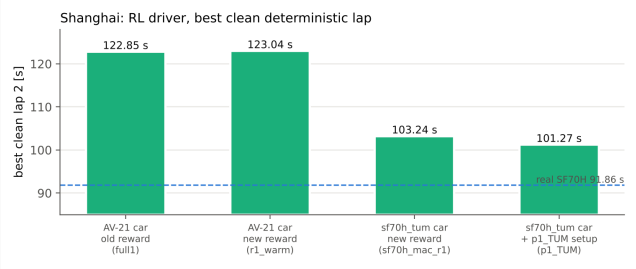
Shanghai: real world vs. QSS vs. our RL driver



No Limebeer lap exists for Shanghai.

- RL **101.27 s**: +9.4 s to Vettel's SF70H (91.86 s), +6.4 s to TUM QSS with no ERS/DRS.
- RL beats our QSS on its own safe line (103.56 s).
- Our QSS on TUM's raceline (95.73 s) is within 0.84 s of TUM.

Shanghai: how the gap closed

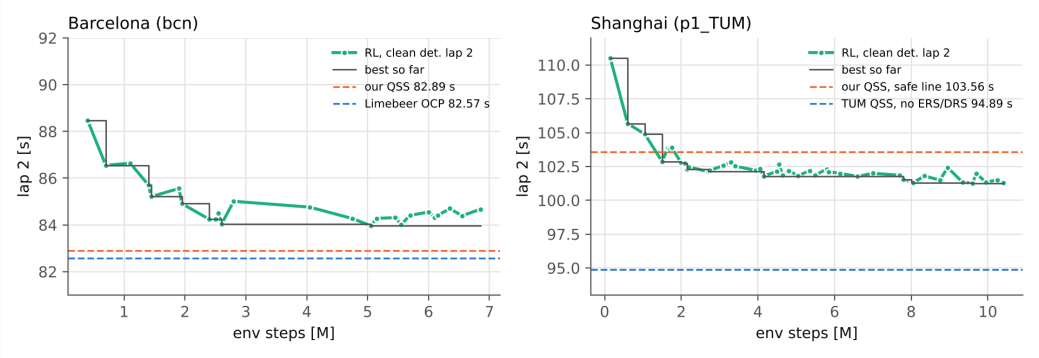


Gap to the real SF70H: **31.0 s** → **9.4 s**.

Remaining gap (Shanghai):

- ERS + DRS off: ≈ 3.6 s (TUM 94.89 → 91.24 s)
- AV-21 tyre transients: up to ≈ 8 s in closed loop. `sf70h_f1` fixes part of it.
- Track check uses the full footprint. TUM's raceline keeps the car on track for only 70 % of the lap.

Training curves (clean deterministic laps only)



Both runs plateau. Barcelona: best 83.97 s, then 84.0–84.7 s with only 26/142 clean deterministic evaluations. Shanghai: 101.3–102 s from 4M to 10.5M steps.

Open points

Summary and next steps

Summary

- The time-optimal reward (no jitter term, linear $|e|$ shaping as PBRS) lets the driver beat the QSS lap on its own reference line.
- The AV-21 tyre slip behaviour explains much of the gap between our car and an F1 car. `sf70h_f1` replaces it with F1-sourced values.
- Our OCP reproduces Limebeer's Barcelona lap within 0.3 %. RL is within 1.7 % of it (with a different car).

Decisions needed

1. Heave damping for `sf70h_f1`, or softer heave springs (porpoising).
2. I_z: keep 1159.6 (recommended) or use 560 kg m^2.
3. RL on `sf70h_f1`: scale env steering to 0.30 rad and re-tune demo traction-control targets.
4. Then: ERS on, a racing line sized for the car's footprint, and seeds 1–2.

Sources in the repository

- Reward: `docs/specs/R1_time_optimal.md`, `docs/research/reward_redesign.md`, `rlsetup/driving/config.py`, DECISIONS #13, #19
- Runs: `docs/results/{D2_full1, R1_warm, sf70h_mac_r1, bcn}`, `comparison/training_stop_20261029_094402Z/p1_TUM`
- AV-21 diagnosis: `comparison/tum_ocd_debug_20261031_144747Z/FINDINGS.md`, `docs/research/f1_car_parameters.md`
- New car: `docs/reports/SF70H_F1_UPGRADE.md`, `docs/results/sf70h_f1/results.json`; ocd_first_lap branch `sf70h-f1`: `ocd_lap/cars.py`, `ocd_lap/aero_map.py`, `docs/SF70H_TUM_CAR.md`, `patches/000{1,2}`
- Tyre curves: `mf52.py` here (Python copy of OCD `lib/ocd_tire_models_cpp/src/mf_52.cpp`)
- OCP: `docs/results/limebeer_ocp/REPORT.md`, `docs/research/limebeer_lap_times.md`
- Real laps: `docs/research/shanghai_lap_records.md`, `shanghai_sim_and_car_reference.md`

Most of this material is on branches `comparison-tum-vs-rl` / `sf70h-f1`, not yet on `main`.